Design Project Report

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Title: 4-Bit Priority Encoder

Theory

A **Priority Encoder** is a combinational logic circuit that encodes the input binary data according to the highest-priority active input. In a 4-bit priority encoder, four input lines are provided (I_0, I_1, I_2, I_3), and the output consists of two encoded bits (Y_1, Y_0) along with a valid bit (V) which indicates if any input is active.

When multiple inputs are active simultaneously, the input with the highest priority (highest index) is encoded as the output.

Truth Table

I0	I1	I2	I3	Y1	Y0	V
0	0	0	0	X	X	0
1	0	0	0	0	0	1
X	1	0	0	0	1	1
X	X	1	0	1	0	1
X	X	X	1	1	1	1

Boolean Equations

From the truth table, the simplified Boolean equations for the outputs are given

$Y_1 = I_2 + I_3$

$Y_0 = I_1\overline{I2} + I_3$

$V = I_0 + I_1 + I_2 + I_3$

NAND Gate Implementation

Using NAND logic, the above equations can be written as:

Y1 = $\overline{\overline{I1}\ \overline{I3}}$

Y0 = $\overline{\overline{I3}\ (\overline{\overline{I2}\ I1})}$

V= $\overline{\overline{I3}\ \overline{I2}\ \overline{I1}\ \overline{I0}}$

These equations form the basis for the NAND gate-based implementation in Cadence Virtuoso.

Implementation in Cadence Virtuoso

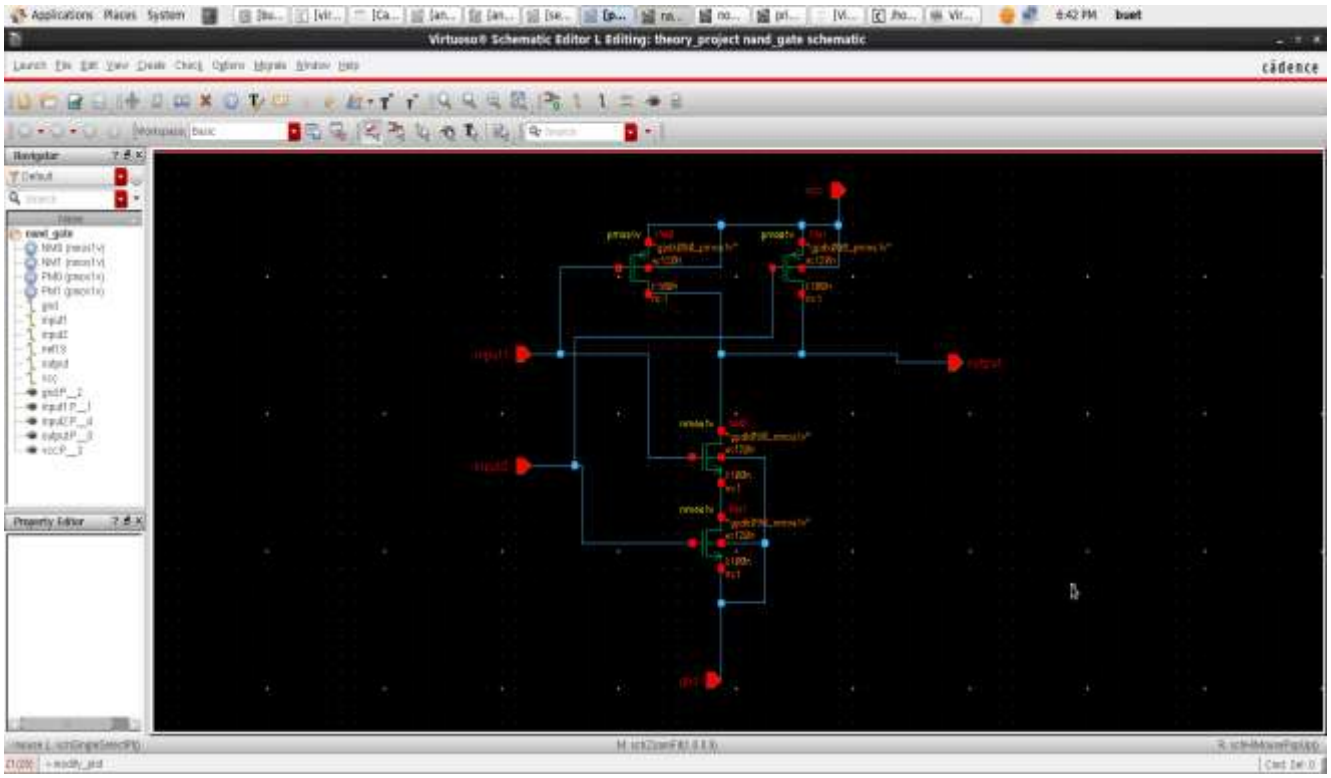
The circuit was implemented in **Cadence Virtuoso** using the **gpdk90nm technology library**. Each NAND gate was first designed and simulated individually before being used to construct the full encoder.

i) NAND2 Gate

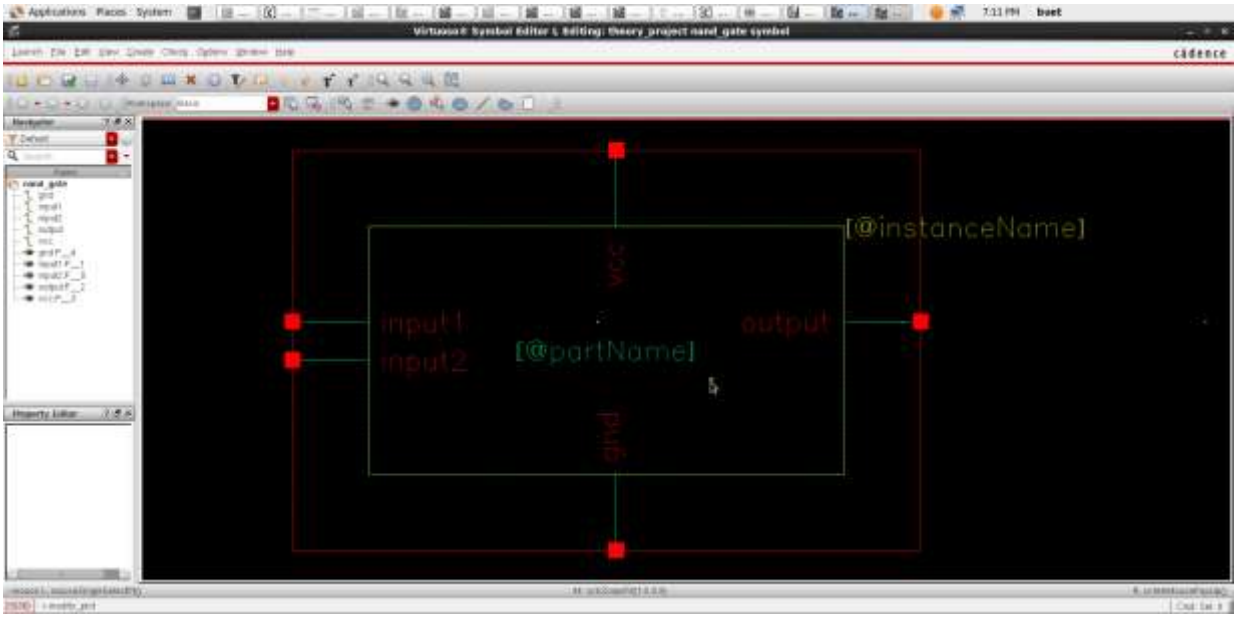
Equation: $Y = \overline{(A \cdot B)}$

The NAND2 gate was designed using two PMOS transistors in parallel and two NMOS transistors in series.

Circuit Diagram:



Symbol View:

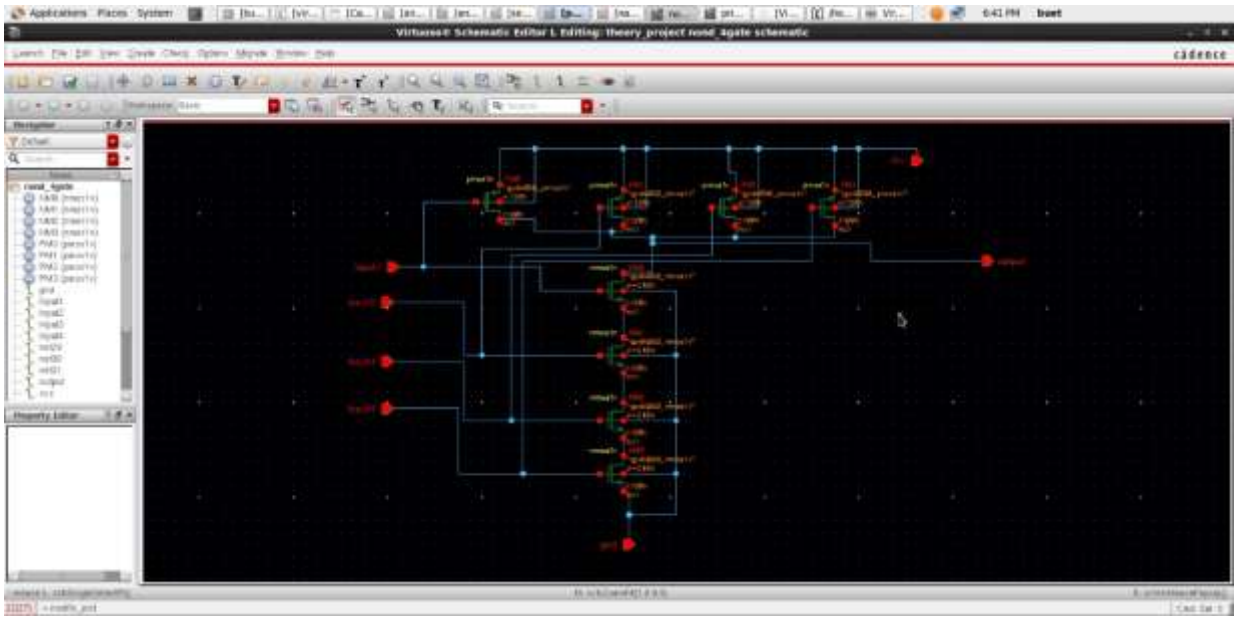


ii) NAND4 Gate

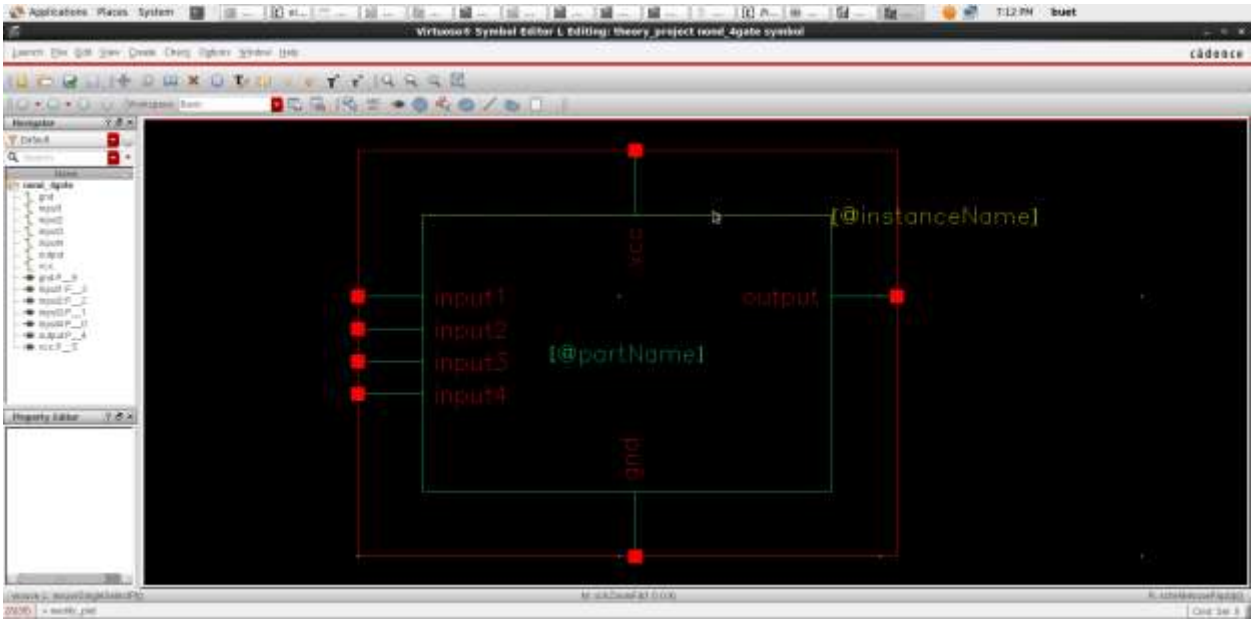
Equation: Y = $\overline{(A \cdot B \cdot C \cdot D)}$

The NAND4 gate was designed using four PMOS transistors in parallel and four NMOS transistors in series.

Circuit Diagram:



Symbol:


[@instanceName]
input1
input2
input3
input4
[@portName]
output

iii) Priority Encoder Circuit

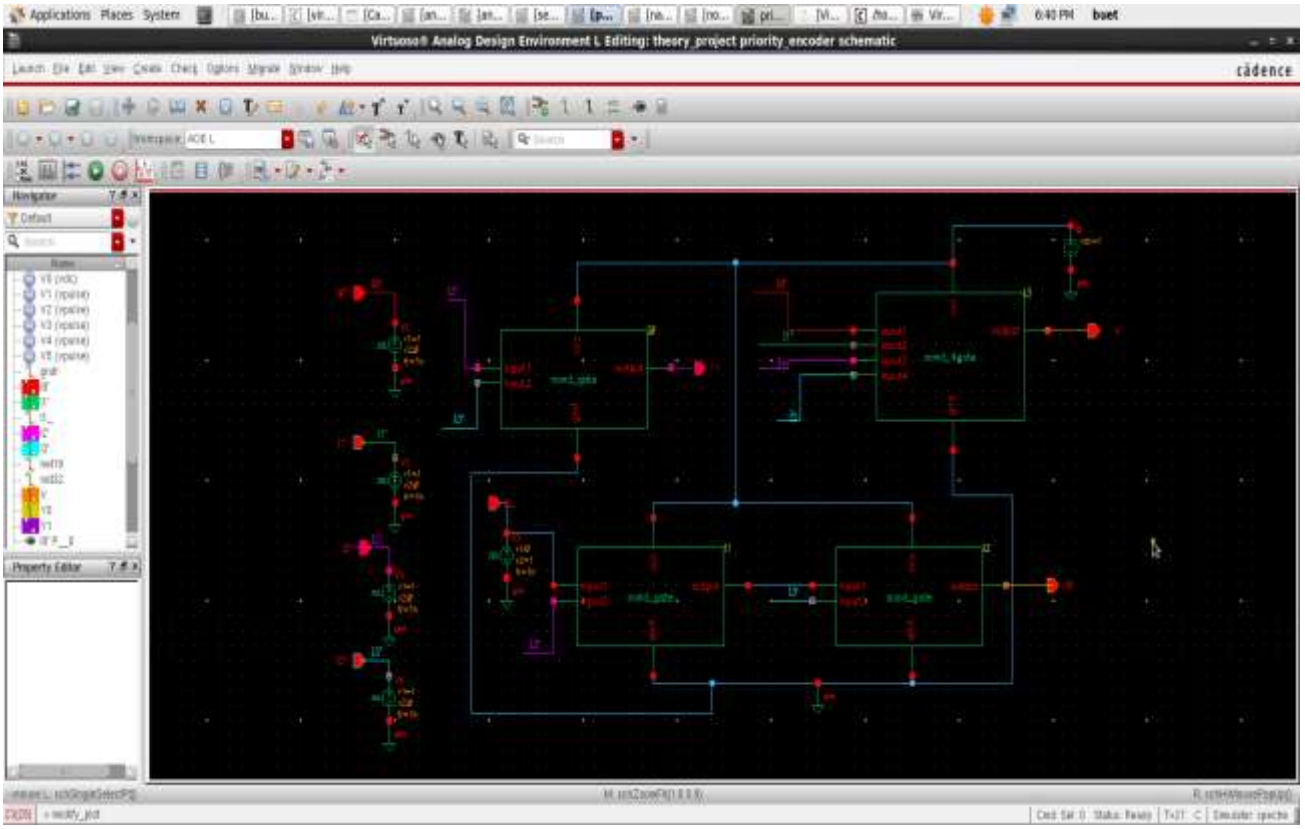
Using the NAND2 and NAND4 cells, the following equations were implemented:

Y1 = $\overline{\overline{I1}\ \overline{I3}}$

Y0 = $\overline{\overline{I3}\ (\overline{\overline{I2}\ I1})}$

V= $\overline{\overline{I3}\ \overline{I2}\ \overline{I1}\ \overline{I0}}$

Circuit Diagram:



Output of Priority Encoder Circuit

The transient analysis was performed using **ADE L** Simulator. The input signals (I_0, I_1, I_2, I_3) were applied using **VPULSE sources**. The simulated waveform below shows the outputs Y_1, Y_0, and V responding correctly according to the truth table.



Result: The 4-bit priority encoder was successfully designed and simulated in Cadence Virtuoso. The output waveforms matched the expected truth table, confirming correct priority encoding where the highest active input is encoded at the output**.**

Inference: The circuit performed as expected, verifying the NAND-based logic design.
Simulation results validated that the encoder gives accurate outputs for all input conditions, proving the design's correctness and reliabilit